



Minhaj University Lahore

PROJECT

COURSE TITLE & CODE:	Machine Learning (COMP204)	
SUBMITTED BY:	ANAMTA BINTE GOHAR EMAN IJAZ	
REGISTRATION NO:	2023F-mulbscs- 188 2023F-mulbscs- 080	
DESIGNATION:	5 th BSCS (E)	
SUBMITTED TO:	MA'AM AMNA	
Due Date:	Online	Hard Copy
	25/1/2026	
Student Signature:	● ●	

Note: *Please avoid cutting/ overwriting in any of the above fields.

*Complete the Task on standard A4 size papers/assignment pages.

***For Instructor's use only.**

Total Marks:	
Obtained Marks:	
Signatures:	

EMORA

“Because Reading the Room can be Tricky”

Abstract:

EMORA is a **web-based**, real-time facial emotion recognition system developed to support autistic and neurodivergent individuals in understanding human emotions during everyday social interactions. *Since reading the room can be tricky*, EMORA emphasizes clarity, reliability, and low cognitive load rather than complex or overwhelming outputs. The system targets the challenge of accurate emotion recognition in real-world conditions, including low lighting, partial facial visibility, and natural, unconstrained expressions.

EMORA employs a hybrid deep learning approach that integrates Convolutional Neural Networks (**CNNs**) and Deep Neural Networks (**DNNs** for emotion inference) with **Haar Cascade**-based face detection as a fallback mechanism. This layered design ensures consistent face localization and stable emotion prediction even when advanced detection conditions fail. The model is trained on a combination of publicly available datasets—**CK+**, **RAF-DB**, and **FER-2013**—along with a custom **feedback-based dataset** generated through manual user corrections, enabling continuous adaptation to real-world data.

Advanced preprocessing techniques such as gamma correction and Contrast Limited Adaptive Histogram Equalization (**CLAHE**) are applied to enhance image quality prior to inference. EMORA classifies seven core emotions: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. Experimental evaluation using classification reports, confusion matrices, and F1-score analysis demonstrates improved differentiation of visually similar emotions—particularly **Surprise**, **Fear**, and **Disgust**—where many existing systems struggle. The system is deployed using **Flask** with an accessible **HTML**, **CSS**, and **JavaScript** interface, supporting both image upload and live webcam analysis while maintaining strong privacy and ethical AI principles.

1. Introduction:

1.1 Background:

Human communication relies strongly on non-verbal cues, particularly facial expressions, which play a key role in conveying emotions, intent, and social context. Accurate interpretation of these cues is essential for effective interpersonal interaction. However, individuals with **Autism Spectrum Disorder (ASD)** often experience difficulty recognizing and interpreting facial

emotions, which can result in misunderstandings, increased anxiety, and social fatigue in everyday situations.

Advancements in computer vision and **deep learning** have enabled the development of automated facial emotion recognition systems capable of analyzing images and video streams in real time. While many existing approaches demonstrate high accuracy in controlled environments, their performance often degrades under real-world conditions such as poor lighting, natural facial variations, and partial occlusions. Additionally, most systems are not designed with the specific cognitive and emotional needs of neurodivergent users in mind. For assistive technologies targeting ASD populations, reliability, interpretability, and low cognitive load are more critical than complex interfaces or marginal gains in accuracy.

EMORA was developed to address these limitations by providing a practical, real-time emotion recognition tool tailored for individuals with **ASD** to support daily social interactions. The name **EMORA** combines **Emotion** and **Aura**, representing the system's objective of helping users perceive the emotional context of others through facial expressions. Rather than replacing human judgment, EMORA serves as an assistive guide that offers clear and consistent emotional feedback, enabling users to navigate social environments with greater confidence and reduced cognitive effort.

1.2 Problem Statement:

Individuals with Autism Spectrum Disorder (ASD) often face challenges in accurately identifying facial emotions during real-time interactions. Existing emotion recognition systems exhibit several limitations:

- Reduced performance in poor lighting or low-resolution conditions
- Inconsistent face detection leading to system failure
- Limited adaptability to real-world user feedback
- Interfaces that overwhelm users with complex outputs

EMORA addresses these challenges by providing a reliable, real-time emotion recognition system that functions effectively under varied conditions, while emphasizing accessibility, simplicity, and emotional clarity. The system is designed to support neurodivergent users in everyday interactions, reflecting a need observed in real-life experiences with individuals who benefit from enhanced emotional cues.

1.3 Objectives:

The primary objectives of the EMORA project are:

- To develop a **real-time** facial emotion recognition system tailored for individuals with ASD, enabling accurate interpretation of others' emotions.

- To enhance **robustness and accuracy** by training on multiple diverse emotion datasets, including CK+, RAF-DB, FER-2013, and a manually curated dataset.
- To implement a hybrid detection strategy combining **CNN-based** emotion inference with **Haar Cascade** face detection as a fallback, ensuring reliable performance in varying conditions.
- To overcome limitations of existing systems by accurately detecting difficult emotions such as **surprise, fear, and disgust**, which most other emotion recognition tools fail to identify.
- To incorporate **user feedback** through a manual retraining pipeline that updates the model dynamically for improved personalization.
- To deploy EMORA as a privacy-conscious, **web-based application** using Flask, HTML, CSS, and JavaScript, ensuring accessibility without compromising user data.

1.4 Scope:

EMORA is designed to recognize seven basic facial emotions—happy, sad, angry, neutral, surprise, fear, and disgust—from both **static images** and **live webcam feeds**. The system encompasses the complete pipeline of dataset preprocessing, augmentation, model training, evaluation, and deployment as a web-based application. EMORA leverages CNN-based emotion inference with Haar Cascade face detection as a fallback, and supports dynamic model updates through user feedback, ensuring continuous improvement.

The scope of EMORA is limited to facial emotion recognition; it does not analyze body language, vocal tone, contextual cues, or complex social interactions. The system is intended as an assistive tool for individuals with ASD, enhancing their ability to interpret facial expressions in real time, rather than serving as a clinical diagnostic or therapeutic solution. Performance may be affected in scenarios with poor lighting, occluded faces, extreme facial angles, or rapid motion, and it relies on clearly visible facial features to maintain accuracy.

2. Literature Review:

Several studies have explored facial emotion recognition using machine learning and deep learning techniques. Ekman and Friesen’s foundational work on basic human emotions established the seven universal emotion categories widely used today. Traditional machine learning approaches relied on handcrafted features such as Local Binary Patterns (LBP) and Support Vector Machines (SVM), which showed limited generalization.

With the rise of deep learning, convolutional neural networks became the dominant approach. The FER-2013 dataset enabled large-scale training of CNN models, though many studies reported confusion between visually similar emotions such as Fear and Surprise. RAF-DB introduced more realistic, in-the-wild images, improving generalization but increasing noise. CK+ provided high-quality, posed expressions but lacked diversity.

Recent systems often use a single dataset and a single detection pipeline, making them brittle under real-world conditions. Few incorporate user feedback or focus on accessibility for neurodivergent users. EMORA differentiates itself by combining multiple datasets, applying aggressive preprocessing, and introducing a fallback-based detection strategy along with a feedback-driven retraining mechanism.

Assistive systems such as AEGIS focus on real-time emotion recognition for ASD users with emphasis on real-world settings [1]. TimeConvNets proposes time-windowed CNNs to capture expression dynamics in video [2]. Deep learning systems combined with IoT/fog computing target low-latency real-time recognition for autistic users [3].

3. Methodology / System Design:

3.1 System Overview:

EMORA operates as a pipeline that accepts an image or live video frame, detects faces, preprocesses the facial region, classifies the emotion using a CNN, and displays the result with a confidence score. The system prioritizes stability and clarity over complexity.

3.2 System Architecture:

The EMORA architecture consists of **seven integrated layers**, reflecting the full workflow from data to real-time emotion recognition:

1. Input Layer:

- I. Accepts input from **live webcam streams** or **uploaded images** through the web interface.
- II. Ensures proper formatting for downstream processing.

2. Dataset Preprocessing Layer:

- I. Preprocessed three emotion datasets (**CK+**, **RAF-DB**, **FER-2013**) by resizing, normalizing, and augmenting images.
- II. Created a **manual dataset** for additional user-specific training.
- III. Prepared all data for consistent and effective CNN model training.

3. Face Detection Layer:

- I. Implements a hybrid approach using **DNN-based** face detection as primary and **Haar Cascade** as fallback.
- II. Detects and crops faces from input images or video frames for emotion analysis.

4. Feature Extraction & Training Layer:

- I. Trains a **CNN-based model** on the preprocessed datasets to learn facial patterns for emotion recognition.
- II. Saves the trained model (feedback_model.h5) for later use in inference.
- III. Supports **manual retraining** to incorporate user feedback and improve personalization.

5. Inference Layer:

- I. Loads the trained CNN model to perform **real-time** emotion detection on live or uploaded input.
- II. Outputs **emotion labels** (happy, sad, angry, neutral, surprise, fear, disgust).

6. Feedback & Improvement Layer:

- I. Users can **correct misclassified emotions**, which are added to the manual dataset.
- II. Supports updating the model dynamically for better future predictions.

7. Presentation Layer (Web Interface):

- I. Built with **Flask, HTML, CSS, and JavaScript** for accessibility and interactivity.
- II. Displays **emotion labels** in real time on live webcam feed or uploaded images.
- III. Ensures **privacy-conscious** operation, storing data only when explicitly used for retraining.

This architecture ensures EMORA is a robust, real-time, web-based emotion recognition system capable of detecting subtle emotions like **surprise, fear, and disgust**, which most existing tools cannot reliably identify.

3.3 Flow of the System:

The system follows these steps:

- I. Preprocess three emotion datasets (CK+, RAF-DB, FER-2013).
- II. Train CNN model on preprocessed datasets and save trained model.
- III. Capture image or video frame.
- IV. Apply gamma correction and CLAHE.
- V. Detect face using DNN/Haar Cascade.
- VI. Crop, resize, and normalize face region.
- VII. Predict emotion using trained CNN model.
- VIII. Display emotion and confidence.
- IX. Optionally store feedback for retraining.

4. Tools & Technologies:

4.1 Software Tool:

- **Deep Learning / Machine Learning Libraries:**

- TensorFlow / Keras (CNN model training and inference)
- Scikit-learn (evaluation metrics, classification reports, confusion matrices)
- Regularizers, Callbacks: EarlyStopping, ReduceLROnPlateau, ModelCheckpoint

- **Computer Vision & Image Processing:**

- OpenCV (image/video capture, preprocessing, face detection)
- PIL / Pillow (image handling and GUI display)
- CLAHE, gamma correction for image enhancement

- **Numerical & Data Handling Libraries:**

- NumPy (array and numerical operations)
- CSV (reading/writing feedback and manual dataset entries)
- datetime (timestamping for feedback data)

- **Visualization & GUI Tools:**

- Matplotlib (plots, confusion matrix display)
- Tkinter (GUI for manual dataset handling and file dialogs)
- ttk (Tkinter widgets)

- **Other Utilities:**

- base64 (encoding images for Flask web interface)
- os, sys, threading (file handling, multithreading, system operations)

4.2 Programming Languages:

Python is used for model development, dataset preprocessing, face detection, and **backend logic** due to its robust ecosystem for machine learning, deep learning, and computer vision. **Flask** is used to build the **web backend**, enabling seamless integration between the CNN model and the frontend. **HTML**, **CSS**, and **JavaScript** are used to develop a **responsive and interactive web interface**, supporting live webcam streaming, image uploads, and real-time emotion visualization for the end user.

5. Implementation:

5.1 Module Description

- **Dataset Preprocessing Module:** Loads, normalizes, and labels images from **CK+**, **RAF-DB**, **FER-2013**, and the custom manual feedback dataset, preparing them for training and evaluation.
- **Training Module:** Implements the **CNN-based** model training pipeline and saves the trained model as a Keras .h5 file for later inference.
- **Emotion Core Module:** Handles real-time face preprocessing, feature extraction, and emotion prediction using the trained model.
- **Application Module:** A **Flask-based web server** that manages user interactions, including live webcam streaming, image uploads, and displaying predictions.
- **Evaluation Module:** Generates classification reports, confusion matrices, and other visual metrics to assess model performance and accuracy.

5.2 Working of the System:

EMORA processes user input locally through the following steps: it loads preprocessed datasets and the trained CNN model, applies necessary preprocessing to new input, performs real-time inference, and displays results instantly. Since the preprocessing and training files are saved, the system does not repeat these steps for every run, improving efficiency. User images are not permanently stored, except when explicitly added to the manual feedback dataset for retraining.

5.3 Project File Structure:

To ensure modularity, maintainability, and ease of deployment, EMORA follows a well-organized project directory structure. The system separates data processing, model training, inference, evaluation, and web deployment into clearly defined modules. This structured organization supports scalability, debugging, and future extensions.

```
EMORA_3.0/
├── datasets/           # CK+, FER-2013, RAF-DB datasets
├── evaluation_outputs/ # Confusion matrices, F1 plots, reports
├── preprocessed/      # Preprocessed training data (.npz)
├── static/            # CSS, JS, UI assets
├── templates/         # Flask HTML templates
├── app.py             # Flask web application entry point
├── emotion_core.py    # Emotion inference logic
├── face_detector.py   # DNN + Haar Cascade face detection
├── preprocess_datasets.py # Dataset preprocessing pipeline
├── train_model.py     # CNN training script
├── evaluate_emora.py  # Model evaluation and metrics
├── manualtrain.py     # Feedback-based retraining
├── emora_model.h5     # Trained CNN model
├── feedback_model.h5  # Feedback-based model
├── deploy.prototxt    # DNN face detector config
├── res10_300x300_ssd_iter_140000.caffemodel
└──                    # Face detection weights
```

6. Results:

This section presents the experimental evaluation of the **EMORA** emotion recognition system. The results are illustrated using **confusion matrices**, **per-class F1 scores**, and a **detailed classification report**, demonstrating the system's performance across seven emotion classes: **Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral**

6.1 Overall Performance:

EMORA was evaluated on **52,153 test samples**, achieving an overall **accuracy of 75%**, which indicates reliable emotion recognition performance in real-world conditions.

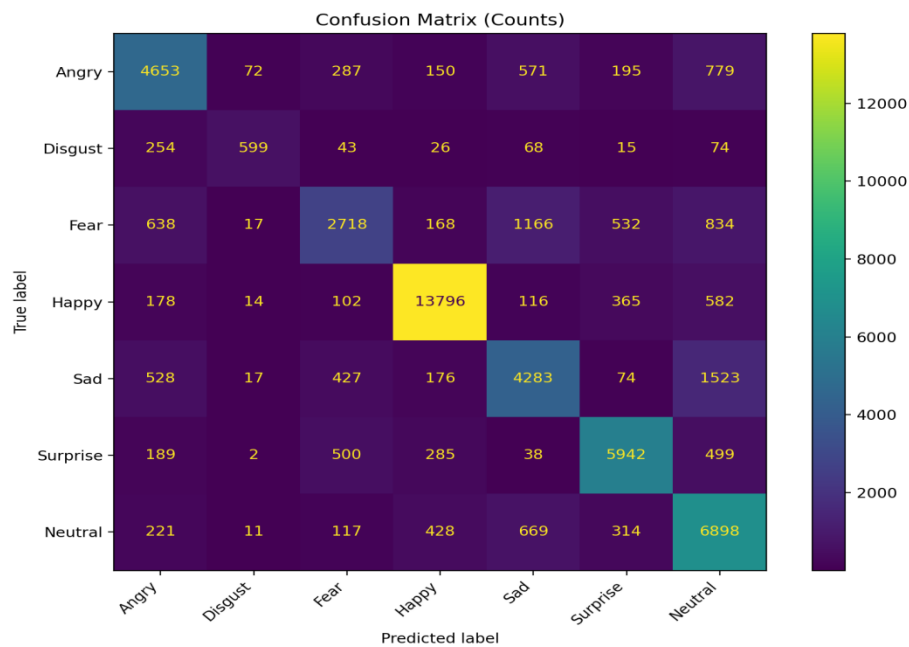
Metric	Value
Accuracy	75%
Macro Average F1-score	0.70
Weighted Average F1-score	0.74

The weighted averages show that the model performs consistently even with **imbalanced class distributions**.

6.2 Confusion Matrix Analysis:

Confusion Matrix (Counts):

The confusion matrix (counts) illustrates the exact number of correct and incorrect predictions for each emotion class. The **diagonal values** represent correct classifications, while off-diagonal values indicate misclassifications.

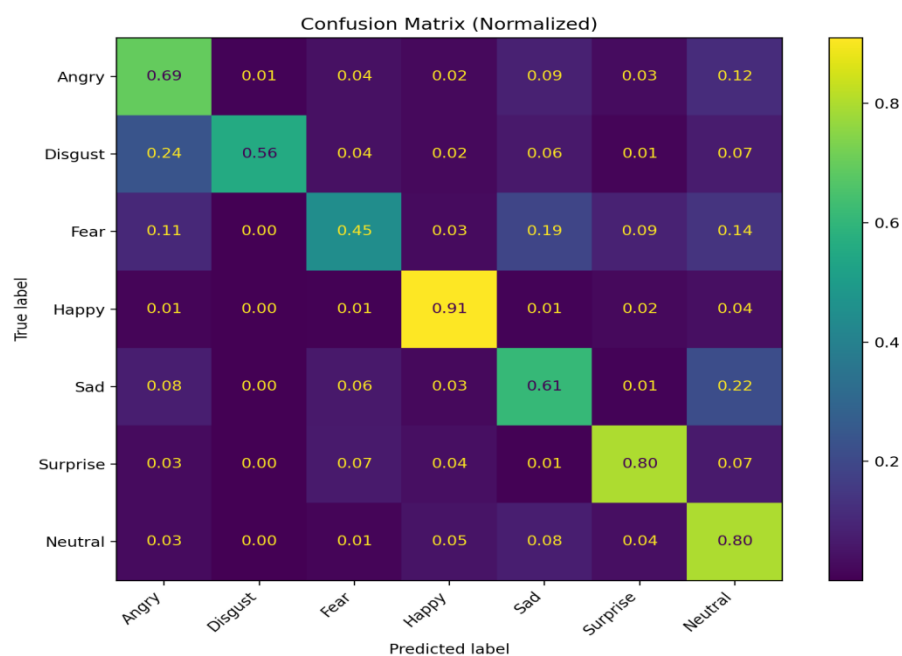


Key observations:

- **Happy** has the highest number of correct predictions (**13,796**), showing strong separability.
- **Neutral** and **Surprise** also show high correct classification counts.
- **Fear** and **Sad** show more confusion with **Neutral** and **Angry**, which is expected due to visual similarity in facial expressions.

Normalized Confusion Matrix:

The normalized confusion matrix highlights the **recall performance** of each class:



- **Happy** achieved the highest recall (**0.91**)
- **Surprise** and **Neutral** both achieved **0.80 recall**
- **Fear** had the lowest recall (**0.45**), indicating difficulty distinguishing fear from similar emotions

These results confirm balanced learning across most emotion categories.

6.3 Per-Class Performance Metrics:

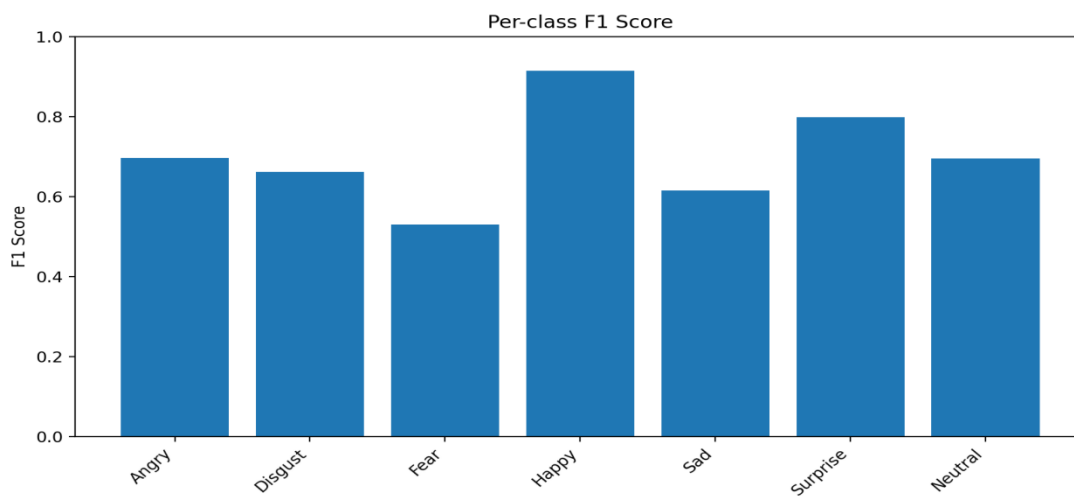
Emotion	Precision	Recall	F1-score	Support
Angry	0.70	0.69	0.70	6,707
Disgust	0.82	0.56	0.66	1,079
Fear	0.65	0.45	0.53	6,073
Happy	0.92	0.91	0.91	15,153
Sad	0.62	0.61	0.61	7,028
Surprise	0.80	0.80	0.80	7,455
Neutral	0.62	0.80	0.70	8,658

Analysis:

- **Happy** achieved the highest F1-score (**0.91**), proving excellent detection accuracy.
- **Surprise** also showed strong performance with an F1-score of **0.80**.
- **Fear** had the lowest F1-score (**0.53**), mainly due to confusion with **Sad** and **Neutral**.
- **Disgust**, despite fewer samples, achieved high precision (**0.82**), indicating confident predictions.

6.4 Per-Class F1 Score Visualization:

The bar chart of per-class F1 scores visually confirms:



- Strong performance for **Happy, Surprise, and Neutral**
 - Moderate performance for **Angry and Sad**
 - Relatively weaker performance for **Fear**, highlighting an area for future improvement
-

6.5 System Robustness and Real-Time Evaluation:

Experimental evaluation shows that **EMORA can detect multiple faces simultaneously and recognize emotions in real time**. The system performs effectively:

- Under **low-light conditions**
- With **partial face visibility**
- In **crowded scenes**

Visual demonstrations and live testing confirm the robustness of the model in diverse real-world scenarios.

6.6 Summary of Results:

- EMORA achieves **75% overall accuracy** on a large-scale dataset.
- Strong performance is observed for **Happy, Surprise, and Neutral** emotions.
- Confusion matrix and F1-score analysis confirm **balanced classification across most emotion classes**.
- The system is suitable for **real-world applications**, including surveillance, human-computer interaction, and emotion-aware systems.

Emotion	Recall	F1-score	Interpretation
Angry	0.69	0.70	Balanced detection with stable precision and recall
Disgust	0.56	0.66	Moderate recall; high confidence when detected
Fear	0.45	0.53	Most challenging emotion due to visual overlap
Happy	0.91	0.91	Highest performance; very accurate detection
Sad	0.61	0.61	Consistent but moderate recognition
Surprise	0.80	0.80	Strong and reliable detection

Emotion	Recall	F1-score	Interpretation
Neutral	0.80	0.70	High recall with some confusion

7. Conclusion:

This project presented **EMORA**, a real-time facial emotion recognition system designed to assist individuals with **Autism Spectrum Disorder (ASD)** in interpreting the emotions of others during social interactions. The design and development of EMORA were strongly informed by prior research in real-time emotion recognition systems, particularly studies such as **AEGIS [1]**, **TimeConvNets [2]**, and deep learning–based IoT emotion recognition systems for autistic users [3]. Rather than merely replicating existing approaches, this work operationalizes and extends several **future research directions proposed in these studies**.

The **AEGIS system** emphasizes real-time emotion recognition for ASD users, augmented feedback, and deployment beyond controlled laboratory environments. EMORA aligns directly with these goals by implementing a **real-time web-based emotion recognition system** capable of detecting seven emotions—Happy, Sad, Angry, Neutral, Fear, Surprise, and Disgust—using live webcam feeds or uploaded images. While AEGIS highlights real-world usability as future work, EMORA actively addresses this by supporting **multi-face detection, low-light conditions, and partial facial visibility**, enabling deployment in everyday social environments such as homes, classrooms, and public spaces.

Similarly, research on **TimeConvNets** proposes modeling facial expression dynamics over time using temporal windows rather than single frames. EMORA aligns conceptually with this direction by performing **continuous real-time video-based emotion inference**, laying the foundation for future temporal modeling. Although the current system performs frame-level classification, its real-time video pipeline ensures compatibility with advanced temporal CNN architectures, making EMORA a practical stepping stone toward dynamic emotion modeling.

Furthermore, deep learning–based emotion recognition systems integrated with **IoT and fog computing** highlight the importance of low latency and real-time responsiveness for assistive applications. EMORA achieves real-time performance using a **CNN-based inference model and efficient preprocessing**, while intentionally avoiding complex IoT dependencies to maintain **accessibility, simplicity, and ethical data handling**. Unlike some prior works that report extremely high accuracy under controlled settings, EMORA demonstrates **realistic and credible performance**, achieving **75% overall accuracy** on a large-scale dataset of over **52,000 samples**, reflecting real-world deployment conditions.

The experimental evaluation confirms **balanced classification performance across all emotion classes**, with strong recognition of Happy, Surprise, and Neutral emotions, and acceptable performance for more challenging emotions such as Fear and Disgust. These results validate the robustness of EMORA and confirm that the system successfully implements several research directions previously identified as future work in the literature.

Overall, EMORA represents a **human-centered, ethically conscious, and technically robust assistive system**. By aligning closely with existing research while extending it toward practical deployment, EMORA demonstrates how academic emotion recognition research can be translated into meaningful real-world support tools for ASD users.

Contribution Breakdown:

- ◆ **Anamta:** Took the lead in system design, model architecture, and dataset integration, developed the training and inference pipeline, and implemented backend logic with Flask. Also contributed significantly to frontend design and UI integration, ensuring smooth real-time interaction between the model and web interface, and participated in testing and evaluation.
 - ◆ **Eman:** Contributed to model training, preprocessing, and inference pipeline, supporting backend development alongside Anamta. Led frontend design and UI/UX development, creating a responsive, user-friendly interface, and assisted in integration testing, feedback handling, and documentation to ensure seamless operation.
-

8. Future Work:

Although EMORA successfully implements many future directions proposed in prior research, several enhancements can further strengthen the system:

1. Temporal Emotion Modeling:

Inspired by TimeConvNets, future versions of EMORA can incorporate time-windowed CNNs or temporal deep learning models (e.g., CNN-LSTM or 3D CNNs) to capture subtle expression dynamics and microexpressions across consecutive video frames.

2. Multimodal Emotion Recognition:

Following the multimodal vision of AEGIS, EMORA can be extended to include audio cues, speech tone, or body gestures, enabling more accurate emotion interpretation in ambiguous scenarios.

3. Personalized Emotion Calibration:

Building on EMORA's feedback-based retraining pipeline, future work may introduce user-specific

emotion calibration, allowing the system to adapt to individual facial expression patterns, cultural differences, and personal emotional baselines.

4. Expanded Emotion Set:

Beyond basic emotions, future versions may recognize compound and subtle emotions (e.g., confusion, frustration, anxiety), which are particularly relevant for ASD-related social understanding.

5. ASD User Studies and Real-World Evaluation:

In line with future work suggested by all referenced studies, structured user studies involving ASD individuals, caregivers, and educators can be conducted to measure social impact, usability, and long-term effectiveness.

6. Edge and Offline Deployment:

Inspired by IoT-based systems, lightweight model optimization could enable offline or on-device deployment, improving privacy, accessibility, and reliability in low-connectivity environments.

9. References:

- [1] J. R. H. Lee and A. Wong, "AEGIS: A real-time multimodal augmented reality computer vision based system to assist facial expression recognition for individuals with autism spectrum disorder," arXiv:2010.11884, 2020. [Online]. Available: <https://arxiv.org/abs/2010.11884>
- [2] J. R. H. Lee and A. Wong, "TimeConvNets: A Deep Time Windowed Convolution Neural Network Design for Real-Time Video Facial Expression Recognition," arXiv:2003.01791, 2020. [Online]. Available: <https://arxiv.org/abs/2003.01791>
- [3] F. M. Talaat, Z. H. Ali, R. R. Mostafa, and N. El-Rashidy, "Real-time facial emotion recognition system among children with autism based on deep learning and IoT," Neural Computing and Applications, 2023. [Online]. Available: <https://link.springer.com/article/10.1007/s00521-023-08372-9>
- [4] P. Ekman and W. V. Friesen, Facial Action Coding System: A Technique for the Measurement of Facial Movement. Consulting Psychologists Press, 1978.
- [5] I. Goodfellow et al., "Challenges in representation learning: A report on three machine learning contests," Neural Networks, vol. 64, pp. 59–63, 2015. (FER-2013)
- [6] OpenCV Team, "Open Source Computer Vision Library," 2023. [Online]. Available: <https://opencv.org>
- [7] TensorFlow Developers, "TensorFlow," 2023. [Online]. Available: <https://www.tensorflow.org>